e)
e) Lipid [illegible]
g) Choleretics (Ja-07, 12)
h) Lactose intolerance (**CMU**: Nov-22, May-20) (Ja-09)
i) Micelle (Ju-09, 11, May-15, Nov-15, Nov-17, 18)
j) Alkaline tide (**CMU**: May-22) (Ju-06,12, 13, 16, Ja-16, May-18, Nov-18)
k) Postprandial alkaline tide (Ja-09, Ju-14)
l) Glucagon like peptide-1 (Ja-11)
m) ANP(Ja-05,06,08,Ju-08)
n) LDH (Ju-14, Ju-15)
o) Intrinsic factor (Ju-14)
p) Composition & function of gastric juice (May-18)
q) What is steatorrhea ? (**CMU**: May-23, 22)

10. List the digestive enzyme with their function. (**CMU**: May-21)
11. List the digestive juice with pH. (**CMU**: May-24)
12. What are the cholenectis? Name them. (**CMU**: May-24)

Renal Biochemistry & Body Fluids

1. Classify body fluid?(**CMU**: May-23, Nov-21) (Ja-13, May-15, 19, Ju-15, 16, Nov-16, 17)
2. Name the regulatory mechanism of body fluid with diagram? & briefly state any of them (**CMU**:May-22, 21) (Ja-09, 07, 08, 11) How ECF is regulated? (Ja-12,13)
3. Mention the different compartments of body fluid with volume. (Ja-14, Ju-14, May-16, Ja-17)
4. Mention the ionic composition of ECF (**CMU**:May-20) & ICF with their normal values. Compare & contrast between ECF & ICF (**CMU**:May-22, 20,19) (Ju-13,14, Ju-15, May-16,17, 19 Nov-16)
5. Name the major electrolyte in ECF with normal range, what is anon gap. (**CMU**: May-22, Nov-21,20)
6. Mention the composition of normal saline. How will you correct Na^+ deficiency in the body? (Ja-07)
7. Give the values of serum electrolytes. (**CMU**:May-23, Ju-14, Ju-15, Nov-17)
   Plasma electrolytes (**CMU**: Nov-22)
8. a) Mention the indicator dilution/ dye dilution technique principal for calculation of body fluid volumes? (**CMU**-May-19, Nov-22) (Ju-13, Ja-16, Nov-16, May-19)
   b) Name the indicators used to measure various body fluid compartments. (**CMU**:Nov-22,May-20) (Ja-06,09, Ju-10,11,12, May-15)
   c) Name the different types of dehydration with eg. Give the composition of ORS (Ju-13)
9. How will you manage/calculate Na^+, K^+ & HCO_3^- deficiency in a patient of 65 kg with Dehydration (Ja-08, Ju-15)
10. How will you calculate osmolarity of plasma? (**CMU**:May-21), (Ju-07, 10)
11. List the plasma changes expected in isotonic dehydration (Nov-16)
12. Give a chart of daily water input & output of the body. Or a chart of water turnover in a day. (Ju-13, May-15, Ju-15)
13. What are the functions of kidney? (**CMU**: May-23,22,19, Nov-20) (May-7, Nov-17, 18)
14. Name the kidney function tests. (**CMU**:Nov-19), (Ja-06, May-15,16, Ju-15) with their reference value (May-18)
15. How will you diagnose a case of Renal failure (**CMU**-Nov-19)
16. Blood Na^+ level 110 mmol/L of body weight 60 kg calculate Na^+ deficiency to correct it. (Ja-09)
17. **Calculate the serum osmolarity:** (Ja-07, Ju-11)

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a) Na^+ =146 mmol/L (Ju-11) b) K^+ =5.4 mmol/L c) Glucose=360 mg/dl.

18. A patient serum K^+ level is 2.5 mEq/l, Body weight 70 kg. How will you calculate to replace K^+ deficiency? (Ju-10) Formula of Na^+ deficit? (Nov-17)
19. A patient of 60 kg has serum HCO_3^- 16.5mmol/L. How will you calculate HCO_3^- deficiency? (Ja-11)
20. A patient serum Na^+ level is 120 mmol/L, body weight is 70 kg. How will you calculate & replace Na^+ deficiency? (Ju-12)
21. Mention the role of kidney in regulation of body water.(Ja-06,10, Ja-16)
22. What is GFR? (**CMU:** May-23, 22, 17). What are the factor that influence GFR?
23. How GFR can be measured.(Ja-02,03)
24. Formula of calculation of Na^+ deficit in plasma in adult male (**CMU:** May-23, Nov-22) (Ju-13)
25. Define GFT with its normal value. (**CMU:**Nov-20)
26. What is metabolic water (Ju-13)
27. Show the calculation of plasma osmolarity (**CMU:** May-22)
28. How will you measure total body water? (Ja-14, May-16)
29. What are the effects of hyponatremia. (**CMU:** Nov-22) Define hyponatremia (**CMU:** May-23) What are the function of Na^+ in our body (**CMU:** Nov-22)
30. Write down the blood picture of respiratory alkalosis. (Ju-14)
31. Write down the formula for calculation of osmolarity of plasma & plasma Na^+ deficit. (Ju-14, Ju-15, May-18)
32. Biochemically how will you diagnos a case of renal failure. (Ja-15, Ju-15)
33. Show with diagram the role of kidney in regulation of acid-base balance. (Ja-15, May-15, 16, 19), (**CMU:**Nov-19)
34. Write in short about the name of abnormal substances in urine & list their detection procedure. (**CMU:** Nov-23, Ja-15)
35. List the major mechanism of regulation of osmolarity of plasma. (Ju-14, May-15) by kidney (Nov-17)
36. Mechanism of kidney in acid base balance with diagram (May-15)
37. How anion gap is calculated & when it is increoned in metabolic condition. (**CMU:**May-22)
38. Why children quickly dehydrated in diarrhoea. (May-15)
39. Osmolarity of normal saline (**CMU**-May-20) (Ju-15)
40. Differentiate between – Absorption and adsorption.
41. List the criteria of dye required for measurement of body fluid volume. (May-16)
42. State briefly with a diagram the mechanism of kidney in maintenance of plasma osmolarity (Nov-16, May-18)
43. Write down the ionic difference between RBC and plasma (Ja-17)
44. What is creatinine clearance rate. (May-16)
45. Give the principle of dye dilution technique. (Ju-16, Nov-17)
46. What are the functions of sodium in the body? Elucidate the hormonal regulation of sodium in kidney. (Ja-17)
47. How kidney convert isotonic glumerular filtrate into concentrated urine? (May-17) (**CMU**-May-19)
48. Give the composition of ECF (May-18, Nov-18)
49. Write down the indicator dilution principle of measurement of body fluid (**CMU**-May-20) (Nov-18)
50. Define GFR (May-19) & give its importance (Nov-18)
51. Give the characteristic of isotonic dehydration (Nov-18)
52. State the necessary parameter required for diagnosis of acid base in balance with their references value.
53. What is bio-chemical picture of dehydration. (**CMU:** Nov-22, May-20)
54. Show the water balance chart. (**CMU:** May-23, 20)
55. Calculate sodium deficit in adult male weighting 65kg. (**CMU**-Nov-20)
57. **SN-** a) Renin angiotensin mechanism (Ja-16, Ju-16)

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b) Renin angiotensin aldosteron mechanism. **(CMU: May-23, Nov-22)**
c) ANP (May-15)
d) Plasma osmolarity (Ja-07, Ju-13)
e) Diruesis. **(CMU:**May-20)
f) ECF serum electrolytes (Ja-11, M-19)
g) Water deficit (May-16, 17)
h) PCO_2 (Ja-15)
i) Renal threshold **(CMU:** Nov-22, May-21)
j) Plasma clearance.
k) Tubular load.
l) TmG value (Nov-17) **(CMU:** May-22)
m) Polyuria (May-15, Nov-16, 17)
n) Renin **(CMU:** Nov-19) (Nov-16, 18)
o) Alkaptonuria (Nov-16)
p) Anion gap **(CMU-**May-21, 20) (May-17)
q) JG cell (May-18)
r) Serum Na^+ **(CMU-**Nov-19) (Nov-18)

58. Classify body fluid compartment **(CMU:**May-21)
59. Discuss the kidney function test. Why serum creatinine is regarded more specific test **(CMU:** Nov-21)
60. What is obligatory urine volume and its importance? **(CMU:** May-23)
61. Infants are very much vulnerable to dehydration related death. How will you evaluate this? **(CMU: May-25)**
62. What are the basic mechanism of urine eostasis formation? what is limiting pH of Urine and why it is called so? How acidic urine forms? **(CMU: May-25, Nov-25)**
63. Mention the factors regulating K+ homeostasis. Write down the cause of hyperkalemia. What are the changes in ECG regarding hyperkalemia. **(CMU: May-25)**
64. write about the body water distribution in 70kg adult man. **(Nov-25)**
65. How water volume is maintained in hypovolemia? Enumerate the cause of hyperkalemia. **(Nov-25)**
66. What are the causes and consequences of hyponatremia? **(Nov-25)**
67. Discuss the role of ANP in water balance. What is water in toxicaion? **(Nov-25)**

SAQ - Paper-II

(Group-A)

Metabolism

Carbohydrate Metabolism

1. **Definition :** a) Glycogenesis b) Gluconeogenesis **(CMU-**Nov-21) (May-15,18 Ju-16)
   c) Glycogenolysis d) Glycolysis
   e) Oxidative phosphorylation (Nov-17)
2. Mention the steps of TCA cycle with ATP calculation **(CMU-**May-22, 21) and sources of acetyl CoA.**(CMU-**Nov-19) Give its importance. (Ju-10,13, Ja-14,15,16,17 May-15)
   Why it is called common metabolic pathway? **(CMU:** Nov-24) (Ju-13, May-15)
3. Give the steps of Glycolysis/ anaerobic glycolysis? **(CMU:** Nov-24)
   Define glycolysis **(CMU:**Nov-24,20) (May-17) with ATP calculation. **(CMU:** May-23, Ju-11,14,15, 16. Nov-15, 16, 18, May-18, Ja-13,16,17) Give it's importance ? (May-17)
4. Show in a diagram the regulation of glucose in blood. (Ja-05) (May-17) **(CMU-**Nov-19)
5. Mention the fates of pyruvate & acetyl-coA? **(CMU-**Nov-20) (Ja-08, Nov-17)
6. Write the steps of glycogenesis. (Ju-06, Nov-17) How it is regulated ? **(CMU:** May-23, 17)
7. Show the steps gluconeogenesis. **(CMU-**May-19), (Ja-07, 08, Ju-13) or Write the substrate, enzyme, site importance of this pathway (Ju-13, May-15, 19)

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8. Importance of HMP shunt. **(CMU:** Nov-22, 20) (Ja-07,14, Ju-07)
9. Write down the fates of glucose after absorption. **(May-25) (CMU:** May-24, 19)
10. Mention the steps of glycogenolysis? What is the importance of Glucose-6 phosphatase enzyme? (Ju-09, Ja-12)
11. Mention important glucose transporters with their function. Illustrate the difference between GLUT-2 & GLUT-4. (Ju-10) Fate of Glucose-6-phosphate. (Ja-12)
12. Difference between liver glycogen & muscle glycogen. **(May-25) (CMU:** May-24) (Ja-02,03,04)
13. Name the inborn errors of metabolism of CHO with their importance (Ju-13)
14. What is lactose intolerance **(CMU:** Nov-23) (Nov-15)
15. State the role of liver in integrated metabolism. (May-16)
16. What is Glucostat ? **CMU**-May-21) (May-17)
17. State the metabolic changes during fasting ? (May-17)
18. State the importance of liver glycogen? (Nov-17)
19. Give the step of conversion of lactate to glucose (May-18).
20. Fate of Lactate (Nov-18)
21. Show with diagram how pentose phosphate pathway protects electrolytes/ erythrocyte against hemolysis. **(May-25) (CMU:** May-24, 20)
22. Show the cori cycle? **(CMU**-May-20). What is its importane.
23. Name the substrate for gloconeogenesis. **(CMU**-Nov-20)
24. **Short notes:** a) Sources & Fates of pyruvates. (Ja-06, 08, Nov-15)
    b) Sources & Fates of Acetyl co-A.**(CMU:**Nov-24)(Ja-15,Ju-16,Nov-18,May-18)
    c) Glucostat. (Ja-06, 11)
    d) Glycogen storage disease. (Ja-07, 06, Ju-09, 11)
    e) Von-Gierke's disease. (Ja-08, 05, 08)
    f) Cori cycle. **(CMU:** May-22, Nov-20, 19) (Ja-08, 14, Nov-16, May-17),
    g) GLUT (Nov-17, 18) h) Pyruvate dehydrogenase complex. (May-17)
    i) PDH complex. **(CMU:** May-22,Nov-21, 20)
25. State the irreversible reaction of glycolysis with enzyme. **(May-25) (CMU:** Nov-23) What are the sources of glucose from noncarbohydrate precursors? **(CMU:** Nov-22)
26. Consequence of glucose-6 phosphate dehydrogene deficiency. **(CMU:** Nov-22)
27. Show with diagram the steps of aerobic glycolysis. **(CMU: Nov-24)**
28. Mention the importance of anaerobic glycolysis. **(CMU: Nov-24)**
29. Give an outline how 38 ATP produce from complete oxydation of one molecule of glucose. **(Nov-25) (CMU:** May-24)
30. How is glycolysis regulated. **(May-25)**
31. Give the calculation of ATP in anerobic glycolysis. **(May-25)**

Protein Metabolism

. Mention the steps of urea cycle with diagram. **(CMU:**Nov-24, 19, May-22,21,19) (Ja-11,13,16, Ju-14,15,16 Nov-15,16, 17, May-19)
. Show in a diagram the steps of amino acid metabolism. Fate of Amino acid. (Ju-15)
. Show in a diagram the metabolism of Fe. What is ammonia intoxication. **(CMU:** May-24, Nov-22) (Ju-10, Ja-12, Nov-15)
Show in a diagram the amino acid pool. **(CMU:** May-22) (Ju-07,10, Ja-17, May-18)
Write in short about post translational modification. **(CMU**-Nov-20)
What are the essential amino acid **(CMU:**May-22)
Write about-
a) PKU **(CMU:**Nov-22, **May-21, 20)**
   Phenylketonuria **(CMU:** Nov-24) **(Ju-05, 06, 09, 10, 13, Ja-12, May-16, 17, 19, Nov-18)**
b) Alkaptonuria (Ja-06, 09, Ju-10)

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c) Glucogenic & ketogenic amino acid.(Ju-04)
7. Mention the disposal of ammonia from body. (Ju-15)
8. Write steps of protein synthesis **(CMU**-Nov-20, 19) (Ja-14, Ja-16, 17, May-16). What is NPN **(CMU**: May-22) (May-17)
9. Source **(CMU**: Nov-20) & fate of amino acid. **(CMU**: May-24, 19) (Ja-14,15, Ju-15)
10. Mention the catabolic pathway of protein metabolism. (Ja-15) How amino acid catabolized in our body **(CMU**: May-23)
11. Soures and way of disposal of ammonia **(CMU**: Nov-24, May-23) (Ju-15, Nov-15, Ja-17)
12. Write the causes and features of ammonia intoxication. **(CMU**: May-23) (Ja-16)
13. Define amino acid pool. Show with a diagram the fates of amino acid nitrogen. (May-16, 17)
14. State the factors promoting elevation of blood lipids. (Nov-17)
15. Write down the fate of ammonia with diagram **(CMU**: Nov-20, May-19) (Nov-18)
16. **Short Notes:**
    a) Nitrogen balance. (Ju-07, 11, 16)
    b) Transamination. **(CMU**: Nov-22, May-20) (Ju-09, Ja-08, May-17, Nov-17)
    c) Oxidative deamination. (Nov-15, Ja-16, May-18) - NPN (May-18)
    d) Amio acid pool. (Ja-13,14)
    e) Source and fates of NH_3. **(CMU**: May-19) (Ju-09,14,16 Ja-12,13, Nov-17, May-19)
    f) Protein turn over **(CMU**: Nov-22)
17. Define amino acid pool. **(CMU**: May-24)

Lipid Metabolism

1. Mention the steps of Oxidation of fatty acid. (Ja-13) Calculate the ATP generated from 16 Carbon fatty acid palmitic acid. **(CMU**: Nov-23, 19) (Ja-15, Ju-15, Nov-15, 18, May-18, 19)
2. Show with a diagram metabolism of VLDL. **(CMU**-Nov-19) (Ju-15, Ja-08,09)
3. Give the major steps of cholesterol synthesis with diagram **(CMU**-Nov-22, 21, 20) and It's function. **(CMU**-Nov-19) (Ju-13, 16 Ja-08, Nov-16, M-19),
4. Mention the steps of ketogenesis. **(May-25) (CMU**-May-22, 20) (Ju-06, Nov-15,18, May-16) Name & fates of the ketone bodies. **(CMU**: May-23,22, Nov-22) (Ja-11,12,13, 14,17 Nov-15)
   Name the factors regulating ketogenesis? (May-17)
5. How prostaglandin are synthesized with diagram? Mention their important function. **(CMU**-May-20, 19) (Ju-09)
6. Show in diagram the steps of synthesis and fate of ketone bodies. (Ju-08, 09, Ja-11,17)
7. Show in a diagram metabolism of HDL. (Ja-13 Ju-11)
8. Write metabolic fate of VLDL with diagram. **(CMU**-May-21, Nov-20).
   How pyruvate convert into glucose.
9. Name the ketone bodies. **(May-25)** What is ketonemia? **(CMU**-May-21) (May-19)
10. Name the lipoproteim. Describe the synthesis and fate of chylomicron. (Ja-11,12,15, Ju-13,15,May-19) What is carnitine transport system?
11. State the intracellular regulation of cholesterol synthesis.
12. Explain with a diagram the mechanism of reverse cholesterol **(May**-19-CMU) transport (May-16) & role of LCAT & CETP. (Ja-11)
13. Calculate ATP production from even/odd number containing amino acid.
14. What are the roles of cholesterol in health & disease (Ju-13) Biomedical importance (May-17)
15. Define lipolysis. Name the lipoprotein. **(CMU**: Nov-19), (Ju-14, Ja-15, Ju-15)
16. Mechanism of ketoacidosis in uncontrolled D.M. (Nov-15) **(May-25)**
17. Show with a diagram the fates of acetoacetate. (May-16, 17)
18. List the factors that reduce plasma cholesterol level. (May-17)
19. Draw the structure of lipoprotein **(CMU**-Nov-20, May-19)

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20. Define lipoprotien. Describe with diagram the metabolic fate of chylomicron. (CMU-May-20)
21. SN: a) β Oxidation. (Ja-08, 14 Ju-11,15 Nov-15, 17, May-19)
    b) Ketosis and ketone body c) Keto acidosis
    d) Lp(a) (Ja-11) - LDL (May-18) Catabolism (CMU: Nov-23)
    e) Lipid profile (CMU-May-21, Nov-20) (Ju-06,12)
    f) Lipolysis (Ju-15) g) HMGCoA (May-17)
    h) Reverse cholesterol transport (CMU- May-22, 20, Nov-21) (May-17)
    i) HMG-COA reductase. (CMU-May-20)
    j) Diabetic Ketoacidosis. (CMU-Nov-20), (Nov-18, May-18)
    k) LDL receptor (CMU-Nov-21)
    l) Carnitine shuttle. (CMU: May-22)
    m) Lipoprotein (CMU: Nov-22)
22. Name some HMG CoA reductase inhibitors (CMU-Nov-21)
23. Define and classify lipoproteins. Give the function of different lipoproteins.(CMU-Nov-21)
24. Give the steps of β Oxidation with ATP calculation. (CMU:May-21)
    Importance of it (CMU: Nov-23)
25. TCA cycle has a dual role of amphibiolic nature-explain it. (Nov-25)

Biological Oxidation

1. SN: a) Oxidative phosphorylation. (CMU: Nov-23, Ju-07, 10, May-16) Example (Nov-25)
   b) Biological oxidation. (CMU: Nov-23, 15, Ja-09)
   c) Respiratory chain d) Oxidoreductase e) ATP (Nov-18) (May-18)
   f) Echosanoids (CMU: May-22, Nov-21)
2. What is respiratory chain? Draw the respiratory chain mentioning (CMU: May-19) the components, sites of ATP production. Name the inhibitor of respiratory chain (CMU-May-19) (Ja-05, 07, 08, 09, 12) Ju-10, Nov-15, 16) & Coenzymes carriers (CMU: Nov-23, May-22)
3. Significance of oxidative phosphorylation. (Ju-07, Nov-15)
4. Write down the steps of respiratory chain (May-18)
5. Write the members of the electron transport chain in order of redox potential & show the steps where ATP is synthesized. (CMU: Nov-20)
6. Define oxidative phosphorylation. Draw the sequence of coenzymes and carries in electron transport chain (CMU: May-21) (Nov-25)

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12. Def... ...Nov-16) ...22) (Ju-13, Ja-17)
13. How vit-A asso... ...and disease. (May-15)
14. What is trace element? **(CMU:**May-22) Write down the source and function of iodine in the body. (Ja-16) importance of iodine in nutrition (Nov-17)
15. Steps of synthesis of vit-D **(Nov-25)** (Ju-15)
16. Write down the role of Zinc, Iodine and iron in heath & diseases **(CMU:**May-23, 22, Nov-19) (May-16)
17. What are the extra ocular features of vitamin A deficiency (Ja-17)
18. What is folate trap mechanism? **(CMU:** Nov-23, Ja-17)
19. Name the factors influence iron absorption (Ja-17)
20. Define & classify trace elements **(CMU:** May-23, 16, Ju-16, Nov-17)
21. **SN:** a) Rickets **(CMU:**May-22) (Ju-06)
    b) Trace elements (Ju-08,11, Ja-11,17, Nov-15, May-19) **(CMU-**May-19)
    c) RDA (Ja-08, Ju-08) d) Beriberi
    e) Micronutrients (Ju-07) Micro-mineral.
    f) Anti oxidant vitamin **(Nov-25)** (Ja-14, May-19)
    g) Scurvy (Nov-15, Ja-15, May-17) h) Pellagra (Ju-16)
    i) 7-dehydrocholesterol **(CMU-**May-19)
    j) Vitamin-K
    k) Vit-A (May-19) l) Hypokalemia **(CMU-**May-21)
22. Name the hemopoitic and energy related vitamins. How Vit-C is associated with post translational modification. **(CMU: May-25)**
23. Write about co-enzyme forms of different vitamins. **(Nov-25)**

DIGESTIVE SYSTEM

Local hormone of GIT

1. List the local hormones of GIT. Give the function of any two of them. (Ja-09, 11, 12, Ju-10)
2. **Discuss source, cause of secretion & function of-**
   a) Gastrin (Ja-09, 08, 07, Ju-16)
   b) CCK-PZ (Ja-08,15)
   c) Local hormones(Ju-12 Ja-13)
   d) Renin (Ja-15)
3. **S/N:** GI hormon. **(CMU-**Nov-20)

Digestive Juice

1. Name the digestive juices? **(CMU:**Nov-22) (Ju-13, Ju-16) with pH? **(May-25)**
2. **Give the composition & function of following:**
   a) Gastric juice. **(CMU:** Nov-22, May-20)) (Ja-07,13, Ju-10, May-15, 19, Nov-15, 18)
   b) Pancreatic juice (Ja-05, 10, 14, Ju-07) **(CMU-**May-19)
   c) Bile (Ja-09, Ju-09)
   d) Succus entericus (Ja-08)

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Stomach & Gastric Juice

1. Show in a diagram the mechanism of HCl secretions in stomach. (CMU:Nov-23, 21) (Ju-16) (Ja-09, 08, 12,13,14, Ju-14, Nov-15, Ja-16)
2. What is achlorhydria? (CMU:Nov-21, May-20)(Ja-12) Function of gastric juices (Cu-10, Ju-
3. Function of gastric juices (Cu-10, Ju-08) Enzymatic constituents of it (Nov-23)
4. Write down the regulatory mechanism of gastric juice secretion. (Ju-06, 10)

Pancreatic Juice

1. Write down the regulatory mechanism of secretion of pancreatic Juice. (Ja-09, 07, Ju-09, 11)
2. Function of pancreatic juice. (Ju-07)
3. State the factors that prevent auto digestion of pancreas. (Ju-11)
4. What is echbolic & hydrolytic type of pancreatic secretion?
5. Name the proteolytic enzymes with corresponding substrate & end products.(Ja-13)
6. Why pancreatic juice is alkaline in nature? (Ja-14)
7. Give the types and composition of exocrine pancreatic juice ? (Ja-16)
8. How proteolytic enzymes are activated **(CMU-May-19)**

Bile

1. Show in a diagram how bile helps in fat digestion & absorption. (Ja-06, Ju-10, May-16)
2. Mention the factors that help in bile secretion. Give the difference between liver bile & bladder bile. (Ja-09, 12)
3. Show in a diagram enterohepatic circulation of bile. (Ja-09, Ju-10, 11, Nov-15)
4. Mention the composition of bile.(May-16) and functions of bile. (May-17, Nov-18)

Gall bladder & liver

1. Function of liver (Ju-07). Mention the liver function test **(CMU-Nov-20)** & their spec indication. (Ja-07,09,10, 12, May-15, Nov-15, 16) **(CMU-May-19)**
2. Classify jaundice? How will you investigate a case of jaundice? (Ja-07, 09)
3. Write biochemical picture of post hepatic Jaundice. **(CMU-Nov-20)**

Digestion and Absorption

1. **Steps of digestion & absorption of-**
   a) Carbohydrate **(CMU:** Nov-23, 21) (Ju-09, Ja-08, 11,13, 14, Nov-15)
   b) Protein (Ju-06, 10, 11,12, Nov-18)
   c) Lipid/fat with diagram (Ja-13,09, 08, 07, Ju-13, Ja-16, Nov-17) **(CMU: May-24,** 19)
2. Show with a diagram how bile helps in fat digestion & absorption. **(CMU:** May-23, Ju-08, Ja-
3. List the digestive enzyme with their site of action for the digestion of CHO, Fat and protein. (Ja-10, 11,12, Ju-11,12)
4. Define digestion, mention the steps of carbohydrate digestion. **(CMU:** Nov-20)
5. List the enzyme of CHO digestion. Name the end product of CHO digestion **(CMU:** Nov-24, May-19). Show in a diagram steps of glucose absorption. **(CMU:** Nov-24) (Ja-11)
6. Name the end product of fat digestion. Show in a diagram **(CMU:** May-19) the digestion absorption of lipid/ **(CMU:** May-22, Nov-20) fat. (Ju-14, 16, Ja-15, May-15, Nov-15, 16, 18)
7. End product of digestion of CHO. Protein, fat **(CMU:** Nov-23, 15, 18)
8. What are the carbohydrate splitting enzyme in GIT. (Ju-16)
9. **SN:** a) CCK-PZ (Ju-08, Nov-17) b) Cholagogues **(Ja-08, 09)**

c) Zymogens (Ja-09)
d) Pancreozymin (Ja-07)
e) Lipid profile (Ju-06,12)
f) Gastrin (Ja-08, 06, Nov-15)
g) Choleretics (Ja-07, 12)
h) Lactose intolerance (CMU: Nov-22, May-20) (Ja-09)
i) Micelle (Ju-09, 11, May-15, Nov-15, Nov-17, 18)
j) Alkaline tide (CMU: May-22) (Ju-06,12, 13, 16, Ja-16, May-18, Nov-18)
k) Postprandial alkaline tide (Ja-09, Ju-14)
l) Glucagon like peptide-1 (Ja-11)
m) ANP(Ja-05,06,08,Ju-08)
n) LDH (Ju-14, Ju-15)
o) Intrinsic factor (Ju-14)
p) Composition & function of gastric juice (May-18)
q) What is steatorrhea ? (CMU: May-23, 22)

10. List the digestive enzyme with their function. (CMU: May-21)
11. List the digestive juice with pH. (CMU: May-24)
12. What are the cholenectis? Name them. (CMU: May-24)

Renal Biochemistry & Body Fluids

1. Classify body fluid?(CMU: May-23, Nov-21) (Ja-13, May-15, 19, Ju-15, 16, Nov-16, 17)
2. Name the regulatory mechanism of body fluid with diagram? & briefly state any of them (CMU:May-22, 21) (Ja-09, 07, 08, 11) How ECF is regulated? (Ja-12,13)
3. Mention the different compartments of body fluid with volume. (Ja-14, Ju-14, May-16, Ja-17)
4. Mention the ionic composition of ECF (CMU:May-20) & ICF with their normal values. Compare & contrast between ECF & ICF (CMU:May-22, 20,19) (Ju-13,14, Ju-15, May-16,17, 19 Nov-16)
5. Name the major electrolyte in ECF with normal range, what is anon gap. (CMU: May-22, Nov-21,20)
6. Mention the composition of normal saline. How will you correct Na^+ deficiency in the body? (Ja-07)
7. Give the values of serum electrolytes. (CMU:May-23, Ju-14, Ju-15, Nov-17)
   Plasma electrolytes (CMU: Nov-22)
8. a) Mention the indicator dilution/ dye dilution technique principal for calculation of body fluid volumes? (CMU-May-19, Nov-22) (Ju-13, Ja-16, Nov-16, May-19)
   b) Name the indicators used to measure various body fluid compartments. (CMU:Nov-22,May-20) (Ja-06,09, Ju-10,11,12, May-15)
   c) Name the different types of dehydration with eg. Give the composition of ORS (Ju-13)
9. How will you manage/calculate Na^+, K^+ & HCO_3^- deficiency in a patient of 65 kg with Dehydration (Ja-08, Ju-15)
10. How will you calculate osmolarity of plasma? (CMU:May-21), (Ju-07, 10)
11. List the plasma changes expected in isotonic dehydration (Nov-16)
12. Give a chart of daily water input & output of the body. Or a chart of water turnover in a day. (Ju-13, May-15, Ju-15)
13. What are the functions of kidney? (CMU: May-23,22,19, Nov-20) (May-7, Nov-17, 18)
14. Name the kidney function tests. (CMU:Nov-19), (Ja-06, May-15,16, Ju-15) with their referenc value (May-18)
15. How will you diagnose a case of Renal failure (CMU-Nov-19)
16. Blood Na^+ level 110 mmol/L of body weight 60 kg calculate Na^+ deficiency to correct it. (Ja-09
17. **Calculate the serum osmolarity:** (Ja-07, Ju-11)

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